Next-Gen 4K AI Camera MaixCAM2 is now crowdfunding! 10X faster than K230 in yolo task! x

Tang Nano 4K

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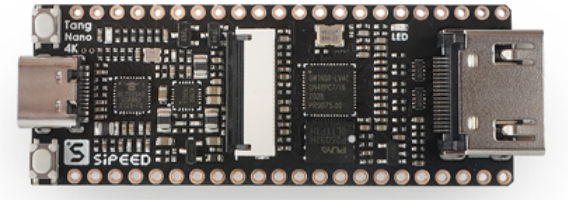
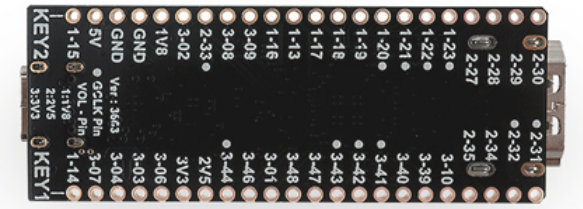
1. Introduction

Tang Nano 4K is a development board designed based on Gowin little-bee GW1NSR-LV4C FPGA chip. The board is equipped with camera interface and HDMI interface. There is also an onboard USG-JTAG debugger, which make it convenient for users to use. Its Cortex-M3 hardcore can help users study mcu.





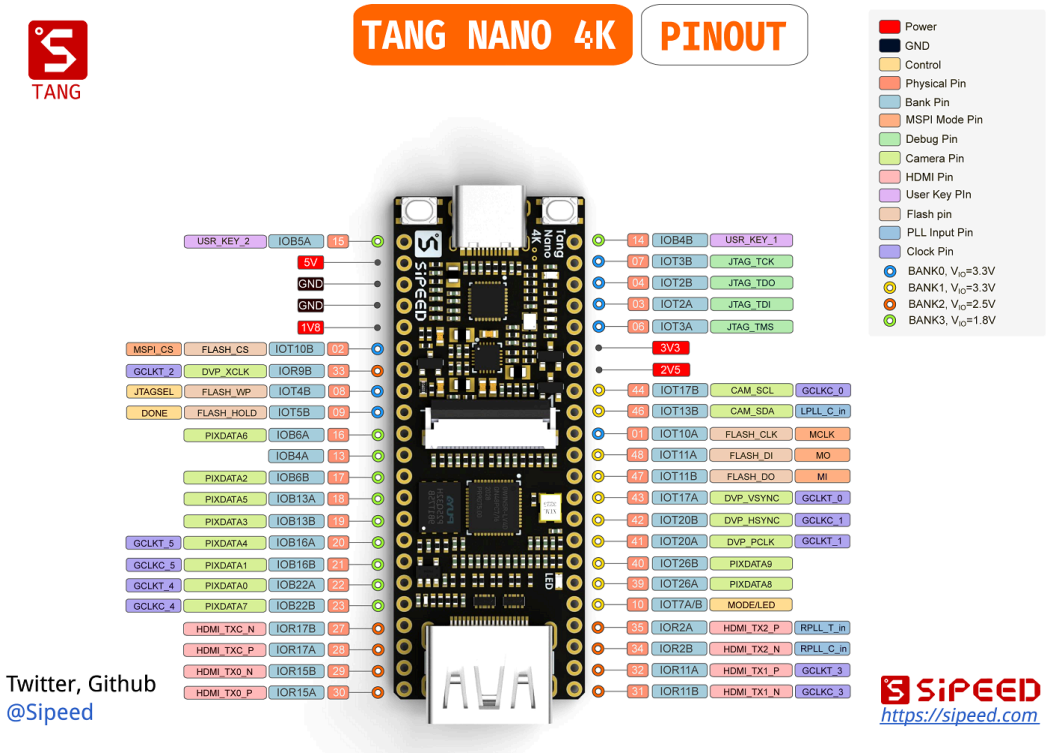
2. Specs

- The sheet below shows difference with previous product

model	Tang Nano	Tang Nano 4K
FPGA chip	GW1N-1-LV	GW1NSR-LV4C
logic units	1152	4608
Register	864	3456
Hard processor	none	ARM Cortex M3
Block SRAM(bits)	72K	180K
User flash(bits)	96K	256K
Number of PLL	1	2
Number of I/O Bank	4	4
Number of users I/O	41	44

model	Tang Nano	Tang Nano 4K
Screen interface	40P RGB LCD interface	HDMI interface
camera interface	None	DVP interface
Size	58.4mm*21.3mm	60mm*22.86mm

2.1. Pinmap



3. Development software

Visit install ide to setup your programming environment.

4. Burn firmware

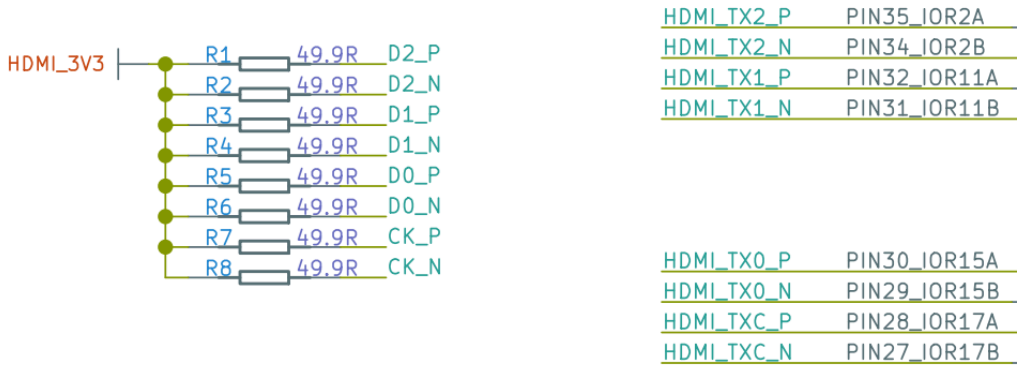
Tang Nano 4K uses the onboard BL702 for jtag, with which to burn bitstream.

Run the Programmer in Gowin IDE to download firmware into FPGA.

5. Informations

- Datasheet
- Schematic
- Bit number map
- Dimensional drawing
- 3D File

- hip Manual
- Examples

6. Addition

1. If you have trouble with this board, you can join our telegram (t.me/sipeed) or contact us on twitter (https://twitter.com/SipeedIO). Leaving message below is also OK.
2. Visit Tang questions first if you have any trouble.
3. Debugging Cortex-M3, we suggest to use serial-port debug way. If you are excellent enough you can try other ways to debug it.
4. THe HDMI ports are multiplexed as IO and routed to the pin headers. The actual results of the IO which are multiplexed with HDMI pins on the pin headers may not be consistent with what you want because of the external pull up.